

Risk-Controlled Tower-Taking in 1v1 MOBA Reinforcement Learning

PPO reward shaping, checkpoint selection, and evidence-driven evaluation for Honor of Kings 1v1

Decision Machine Learning Project | Risk-aware PPO training analysis | Shanghai Innovation Institute

Abstract. This work investigates reward shaping and evidence-driven checkpoint selection for a PPO agent in a long-horizon 1v1 MOBA task. The proposed risk-aware training run introduces terminal outcome rewards, explicit tower-pressure terms, push-window diagnostics, and unsafe-dive penalties. Results show a substantial reduction in death rate, while tower conversion still lags behind the previous peak baseline. We therefore identify step-15003 as the strongest candidate for full matchup-matrix evaluation.

Research Problem

Goal. Train a PPO agent to turn local lane advantages into stable tower-taking wins in a long-horizon 1v1 MOBA environment.

Core difficulty. Immediate combat rewards can make the policy fight, trade, or dive, while the real objective is delayed: preserve enough safety to destroy the enemy tower.

Hypothesis. Terminal outcome reward, explicit tower-pressure terms, and tactical-window diagnostics should improve objective conversion. Unsafe-dive penalties should reduce death inflation without making the policy passive.

Implemented Method

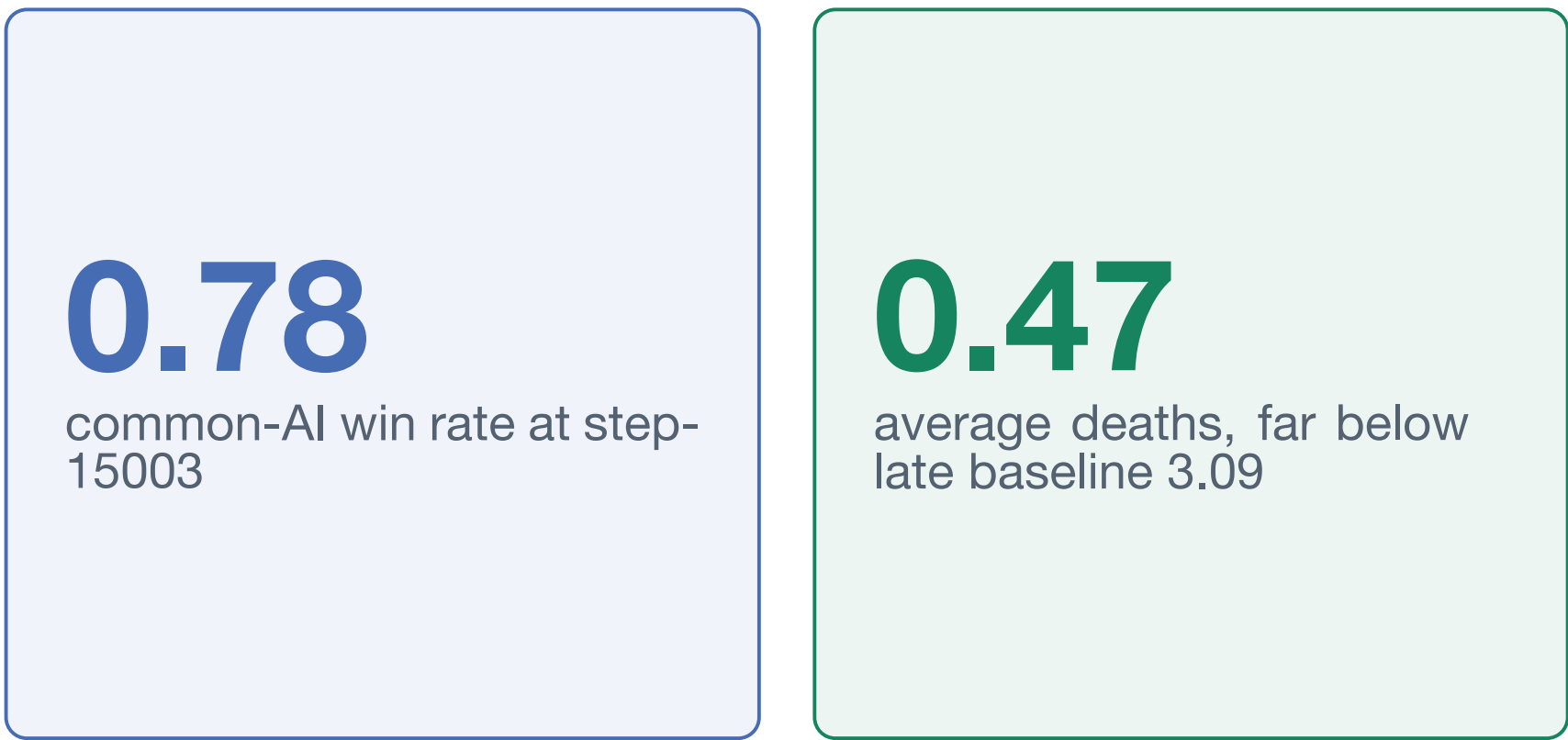
- **PPO backbone:** clipped policy loss, value loss, entropy regularization, GAE, legal-action masking, and gradient clipping.
- **Tower objective:** enemy tower damage, self tower damage, tower destroy, terminal win/loss, and time-out tower-gap rewards.
- **Tactical window:** push-window tower-damage reward plus zero-weight diagnostics.
- **Risk control:** death, unsafe-dive, and unsafe-dive severity penalties.
- **Evidence chain:** checkpoint, lineup, matchup, opponent source, terminal outcome, and reward-detail logging.

$$R_t = \sum_i w_i \Delta f_i(s_t) + 20R_{\text{win}} + 8R_{\text{tower gap}}$$

Acceptance Gates

Gate	Target
Win rate	Beat 0.84, or match it with much lower death
Tower pressure	Enemy tower HP below 1400.59
Death control	Average death below 3.09
Coverage	All 9 hero matchups with side swaps
Diagnostics	Push-window share and unsafe-dive correlation

Best Candidate



Checkpoint ranking selected step-15003. It is the only checkpoint with both high win rate and strong objective pressure. Later checkpoints improve scalar reward but regress on gameplay metrics.

Checkpoint Ranking

Step	Win	Enemy HP	Death	Score
15003	0.78	2129	0.47	75.79
17617	0.53	4106	0.53	48.50
15977	0.53	3978	0.69	48.35
10004	0.53	4735	0.59	47.66
17022	0.47	5423	0.56	41.09

Selection rule. Do not pick the final checkpoint by default. Use win rate, tower HP, death, and matchup evidence.

Candidate vs. Peak Baseline

Metric	Candidate	Baseline	Delta
Win rate	0.78	0.84	-0.06
Enemy tower HP	2129	1401	+729
Self tower HP	4296	7745	-3449
Frame	15726	13670	+2055
Kill	3.44	3.75	-0.31
Death	0.47	2.72	-2.25
Money/frame	0.53	0.58	-0.05
Hero damage	1.63	1.88	-0.25

Interpretation. Safety improved dramatically, but objective conversion is weaker and slower.

Training Dynamics

- **0-2979:** cold start; enemy tower HP remains above 9500.
- **4028-7018:** first breakthrough; win rate reaches 0.47.
- **7992-13964:** plateau around 0.38-0.53 win rate.
- **15003:** narrow peak with 0.78 win rate and 2129 enemy tower HP.
- **15977-17617:** reward improves, but gameplay metrics regress.

Win rate tracks tower pressure, self-tower survival, economy, kills, and outgoing hero damage. Death is no longer the main bottleneck because the risk-aware run has already compressed deaths into a low range.

Candidate Gate

PASS	death control; hero-damage balance; explicit checkpoint selection
WARN	win rate below 0.84; tower pressure below the peak baseline
MISSING	9-matchup matrix, worst-matchup gap, push-window share, unsafe-dive correlation, death p90, self-tower p10, timeout rate

Meaning. The gate failure is actionable: run the missing evidence package before making a final model claim.

Next Experiment

- Freeze **step-15003** as the candidate checkpoint.
- Run 9 hero matchups, side swaps, 20 repeats, and the planned skill grid.
- Preserve reward-detail logs for push-window and unsafe-dive analysis.
- Keep risk shaping, but increase tower-damage and terminal-win alignment.
- Compare full reward, no-window, no-terminal, and death-only-risk ablations.

What improved

Deaths dropped from the late-training baseline 3.09 to 0.47 at the selected checkpoint. The agent learned safety.

What is missing

Enemy tower HP remains 2129, worse than the peak baseline. Safety is not yet converted into tower destruction.

What proves it

The full 9-matchup matrix with push-window and unsafe-dive evidence decides whether the candidate generalizes.

Takeaway

Risk-aware reward shaping can sharply reduce deaths, but stable MOBA success still depends on converting safety into tower destruction. The next decisive evidence is the full matchup matrix for step-15003.